

DESIGNING AN EBAY-LIKE DATA WAREHOUSE

“Fact Finders' Blueprint for Business Intelligence”

Course: IST 722 Data Warehouse Project

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Project Scope

- The **Data Warehouse** project aims to design and implement a **warehouse** that replicates key business processes in an **auction-based online marketplace** similar to **eBay**.
- The project will integrate and analyze **bid transactions, item listings, auction statuses, and user behaviors** to provide meaningful insights using **Business Intelligence (BI)** tools like **Power BI**.
- **The scope includes:**
 - **Extracting, Transforming, and Loading (ETL) data** from operational vBay tables (vb_bids, vb_items, etc)
 - Designing a **star schema** to model auction transactions, item listings, and user bidding activities.
 - Implementing a **dimensional data warehouse** using fact and dimension tables.
 - Building **Power BI dashboards** for analyzing auctions, bidding trends, and user behaviors.

DATASET - OVERVIEW

- The **vBay dataset** represents an **online auction platform**, similar to **eBay**, where users can list items for bidding and place bids on available items. It contains essential tables that track **bidding transactions, item listings, item categories, and bid statuses**.

Table Name	Description
<code>vb_bids</code>	Stores all bid transactions, including bidder details, item ID, bid amount, and bid time.
<code>vb_items</code>	Contains item listings, including item descriptions, starting prices, seller details, and auction durations.
<code>vb_item_types_lookup</code>	A lookup table for categorizing items (e.g., Electronics, Clothing, Collectibles).
<code>vb_bid_status_lookup</code>	A lookup table that defines the status of bids (e.g., Active, Won, Lost).

PROJECT OBJECTIVE

- **Tracking bidding activity** – Analyze the number of bids per item and highest bid trends.
- **Auction success rate analysis** – Identify items that received bids and met their reserve price.
- **Popular item categories** – Determine which item types attract the most bids.
- **User behavior analysis** – Understand bidder engagement and frequency of bids.

BUS MATRIX OVERVIEW

Business Process Name	Fact Table	Fact Grain Type	Granularity	Facts	Dimension Tables			
					dim_bidder	dim_item	dim_status	dim_time
Bidding	fact_bids	Transaction	One row per bid placed	bid_amount, bid_time	x	x	x	x
Item Listing	fact_items	Accumulating Snapshot	One row per listed item	listing_price, final_price, num_bids		x		x
Auction Complete	fact_auction_status	Periodic Snapshot	winning_bid, num_bidders, auction_duration	inventorylevel, reordercount	x	x	x	x

FACT TABLE AND DIMENSION TABLE

Detail each fact table along with its granularity and the metrics it captures:

- **fact_bids:** Transactions, captures bid amount, bid time.
- **fact_items:** Accumulating Snapshot, captures listing price, final price, number of bids.
- **fact_auction_status:** Periodic Snapshot, captures inventory level, reorder count.
- **The dimensions:** dim_bidder, dim_item, dim_time.

CONCLUSION

This **Warehouse** project successfully models an **eBay-like auction platform**, enabling businesses to analyze **bidding behaviors, auction effectiveness, and item popularity**. Through **Power BI dashboards**, stakeholders gain valuable insights into:

- The **most competitive bidding categories**.
- The **conversion rate of auctions** (items that meet reserve price vs. unsold).
- The **activity of top bidders and sellers**.